The Clebsch hexagon code: rigidity from a conic deep-hole locus

Tavis Rudd

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Abstract

Among the six-arcs of $\mathrm{PG}(2, 11)$, exactly one projective class has the maximum-distance syndrome directions of its associated MDS code contained in a conic: the Clebsch hexagon. This coding-theoretic condition recovers its projective stabilizer A_5, and the same class is characterized by containment in a plane curve of degree at most three. The arc itself lies on no conic, so its $[6, 3, 4]_{11}$ MDS code is non-GRS. Its covering radius is three, and its twelve projective deep-hole directions are exactly the $\mathbb{F}_{11}$-points of a conic. A quadratic syndrome test is a complete distance oracle; its ambiguity data recover the Brianchon points, self-polar triangles, and an invariant support chirality. A line bound, chord identities, and the Sylvester graph isolate $q = 11$ among six-arcs; for arc sizes four through seven, the only other case is a four-frame in $\mathrm{PG}(2, 5)$. The two exceptional secant arrangements are the finite reductions of the reflection arrangements H_3 and A_3.

Keywords. Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic; reflection arrangement.

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1 Introduction

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H. For a received word v, write $\sigma(v) = Hv^\mathsf{T}$ and

$$w(s) = \min\{\mathrm{wt}(e) : He^\mathsf{T} = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ. The syndrome s specifies one coset of C, whose coset leaders are its weight-$w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\mathrm{proj}}(C) = \{[s] \in \mathrm{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q - 1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n - k$, the parity-check columns form a projective arc in $\mathrm{PG}(r - 1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r - 1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write $\mathcal{U}(A)$ for this extension locus. If $\mathcal{U}(A) \neq \emptyset$, then the associated code has covering radius three and $\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C)$. This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [10]; their Theorem 6.3 supplies the leader count used in Section 3. See also Hirschfeld [17] for the plane arc/covering-radius setting.

The three correspondences used throughout are

$$\begin{array}{rcl} \text{parity-check column ray} & \longleftrightarrow & \text{arc point,} \\ \text{non-arc point } x \text{ on } i \text{ secants} & \longleftrightarrow & i \text{ weight-two leaders per syndrome above } x, \\ \text{uncovered point} & \longleftrightarrow & \text{projective deep-hole direction.} \end{array}$$

The paper studies one small exception to the GRS picture: a non-GRS $[6,3,4]_{11}$ MDS code whose projective deep-hole locus is exactly the $\mathbb{F}_{11}$-point set of a nonsingular conic. Its parity-check columns form the *Clebsch hexagon*, a six-arc in $\mathrm{PG}(2,11)$. Thus a natural property of the covering-radius structure, with no assumed symmetry or conic construction, singles out a classical configuration.

The main theorem starts from no symmetry or conic construction. Among all six-arcs of $\mathrm{PG}(2,11)$, conic containment of the projective deep-hole syndrome locus reconstructs the Clebsch hexagon and its A_5 stabilizer. A line bound, a chord-defect identity, and Dye's Brianchon theorem give the conceptual implication; the census is needed only for the numerical clause. We then prove quantitative gaps, reconstruct the Brianchon geometry from decoder ambiguity, and isolate $q = 11$. For a Clebsch hexagon K, the same chord identity gives $|\mathcal{U}(K)| = q^2 - 14q + 45$, with off-conic excess $(q-4)(q-11)$ when $q \equiv 3 \pmod 4$. A reflection-arrangement model organizes this factorization and the full decoder ambiguity spectrum: the fifteen chords are reduced projectivized H_3 mirrors. The six joins of a four-frame are similarly the projectivized A_3 mirrors, placing the two conic-filling cases in one framework; separate geometry identifies the two complements as conics.

The configuration goes back to Clebsch [5, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [6, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [15, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [22]. Deep holes of Reed–Solomon codes have both an algorithmic history [16, 9] and a geometric classification in the projective case [23]. Those results concern GRS codes, whose arcs lie on normal rational curves; the code here is not GRS.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [5, 6, 15, 2, 22]. Sadeh classified the six-arcs of $\mathrm{PG}(2,11)$ [20]; we claim no priority for that census or its extension counts. The companion arcs paper records the single conic-locus identification of Proposition 3.2, with a kernel-checked certificate [8, Prop. 8.2]; we reprove it for self-containment, and no result below depends on the companion paper. The new contribution is the symmetry-free coding criterion and its consequences: low-degree rigidity, quantitative gaps, leader chirality, decoder reconstruction, and the finite-field boundary. Edge already obtains the real Clebsch hexagons from the six opposite-vertex axes of an icosahedron [6, §§19–20]; the icosahedral mirror arrangement and its $6_5, 10_3, 15_2$ singularity ledger are likewise classical [4]. We claim no priority for this icosahedral geometry or its H_3 name. The finite-field characteristic-polynomial method is classical [1], and Jurrius and Pellikaan already derive coset-leader and list-weight enumerators from the intersection lattice of the parity-check derived arrangement, explicitly including redundancy-three MDS secant arrangements [18, Prop. 3.11, Thms. 5.3 and 5.7, Ex. 5.10]. Our paper-specific additions are the explicit $\mathrm{PGL}_3(\mathbb{F}_{11})$ bridge from the reduced H_3 fivefold points to the displayed code, its specialization to this code's decoder strata, and the joint A_3/H_3 organization of the two conic-filling exceptions.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}:\ XZ = Y^2$$

in $\mathrm{PG}(2,11)$, with its 12 points identified with $\mathrm{PG}(1,11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3,11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

A convenient representative is

$$A = \{(1,10,0),\ (1,9,1),\ (1,4,7),\ (1,8,5),\ (0,1,4),\ (1,1,7)\}.$$

It is a six-arc disjoint from $\mathcal{C}$; its fifteen joins also avoid $\mathcal{C}$. The following invariant construction explains its symmetry.

Definition 2.1 (Clebsch hexagon)**.** Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The six Sylow 5-subgroups of A_5 each fix two points of $\mathcal{C}$. The *Clebsch hexagon* is the set of poles of the six chords joining these fixed pairs.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an external, or *passant*, line is disjoint from the conic. Edge's six vertices are external, their fifteen joins are external, and those joins concur three at a time at ten internal Brianchon points [6, §§29–32]. Thus the hexagon is a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [2, pp. 143,146]. Their $q = 11$ list also contains a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. Thus exteriority alone gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; the arc hypothesis selects the Clebsch branch, and Proposition 3.2 proves equality.

This is the arc K_2 of Storme and Van Maldeghem [22, Props. 11–12] and Dye's synthetic hexagon when 5 is a square in the base field [15, Theorem 1]. Dye proves uniqueness, stabilizer A_5, and the unique associated polarity; in the displayed coordinates its conic is $\mathcal{C}$. His calculation also shows that the six vertices lie on no conic in characteristic 11 [15, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem record that the arc is incomplete at $q = 11$ [22, Prop. 13], so the associated redundancy-three code has covering radius three.

3 The code and its projective syndrome locus

Form the 3×6 matrix H whose columns are (representatives of) the six points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6,3,4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [21].

The arc–coset dictionary from the introduction applies without a GRS hypothesis [10, Def. 6.1, Thm. 6.3]. Each uncovered direction x lifts to $q - 1$ weight-three cosets, and each has $\binom{6}{3} = 20$ minimum-weight leaders. Geometrically, the same condition says that adjoining x as a seventh column preserves the MDS property.

The stabilizer already forces the small point sets that will occur.

Proposition 3.1 (The A_5 point orbits). *Let $G = \mathrm{Stab}(A) \cong A_5$. Its orbits on* $\mathrm{PG}(2, 11)$ *have lengths*

$$6,\ 10,\ 12,\ 15,\ 30,\ 30,\ 30.$$

The first four are respectively A, the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G-orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [15, Theorems 2–3 and Corollary 1, pp. 276–279]. We derive the orbit sizes without using the finite enumeration.

The argument first determines the points with each possible nontrivial stabilizer, then applies orbit–stabilizer to recover the orbit lengths.

Since $11 \nmid 60$, the representation is semisimple, and the ternary icosahedral representation is irreducible; hence G fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in $\mathbb{F}_{11}$, shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in $\mathbb{F}_{11}$; and the three order-five eigenvalues are distinct because $5 \mid 10$. Restricting the same representation to the standard subgroups of A_5 gives the following common-fixed-point ledger; the standard subgroup classification of A_5 shows that these are all possible nontrivial proper point stabilizers. Here D_5 has order ten, and the last column counts points whose stabilizer is exactly the displayed subgroup type.

H	A_4	D_5	S_3	C_5	V_4	C_3
number of conjugate subgroups	5	6	10	6	5	10
fixed points for one H	0	1	1	3	3	1
points with stabilizer conjugate to H	0	6	10	12	15	0

By orbit–stabilizer, the third row already gives the orbits of lengths 6, 10, 12, and 15; only points with exact stabilizer C_2 remain. To spell out the subtraction in the last row: the normalizer D_5 fixes one of the three C_5 eigenlines and interchanges the other two, giving 6 points with stabilizer D_5 and 12 with stabilizer C_5; the unique C_3-fixed line is already fixed by its S_3 normalizer; and A_4 cycles the three V_4 eigenlines, leaving all fifteen with exact stabilizer V_4.

It remains to account for exact C_2 stabilizers. Each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4. Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2. There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6 + 10 + 12 + 15 + 90 = 133$ exhausts the plane. Dye's Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5-stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [15, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. □

Proposition 3.2 (The projective deep-hole syndrome locus is the conic). *The covering radius of C is three, and its projective deep-hole syndrome locus $\mathcal{D}_{\mathrm{proj}}(C)$ — the set of directions $x \notin A$ on no secant of A — is exactly the twelve points of $\mathcal{C}$.*

Proof. The set $\mathcal{U}(A)$ is G-invariant. Dye's Clebsch classification gives exactly ten Brianchon points. The relevant double count is short. The fifteen edges contribute 150 off-arc incidences. The 45 pairs of disjoint edges meet once, so if n_i counts off-arc points lying on exactly i edges, then a point on three edges accounts for $\binom{3}{2} = 3$ such pairs and

$$n_1 + 2n_2 + 3n_3 = 150, \qquad n_2 + 3n_3 = 45, \qquad n_3 = 10.$$

Thus the edges cover $n_1+n_2+n_3 = 115$ of the 127 off-arc points, leaving $|\mathcal{U}(A)| = 12$. This is the $q = 11$ specialization of the chord-defect identity in Lemma 6.1. By definition $\mathcal{U}(A)$ is disjoint from the unique six-point orbit A. The orbit lengths in Proposition 3.1 therefore force this invariant twelve-set to be the unique twelve-point orbit $\mathcal{C}(\mathbb{F}_{11})$. By the dictionary above this is exactly the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three rather than two. Dye's edge criterion and the complete-exterior-set description of Blokhuis–Seress–Wilbrink independently supply the classical inclusion $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$ [15, discussion preceding Theorem 6, p. 281]; see also [2, pp. 143,146]. $\square$

Corollary 3.3 (Directions, cosets, leaders, and received words)**.** *The projective deep-hole syndrome locus has* 12 *directions, exactly the points of* $\mathcal{C}$*. Above them lie* $12(11-1) = 120$ *nonzero maximum-distance syndromes, hence* 120 *deep-hole cosets. Each such coset has* $\binom{6}{3} = 20$ *minimum-weight leaders, for* 2400 *leaders in all, and contains* $|C| = 11^3 = 1331$ *received words. Consequently* C *has* $120 \cdot 1331 = 159720$ *received-word deep holes.*

Proof. Immediate from Proposition 3.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \to \mathrm{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. $\square$

3.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^{\times} \longrightarrow \mathrm{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars. The sequence splits. Since $\gcd(3, 10) = 1$, each projective class has a unique determinant-one representative; their monomial lifts give an A_5 complement. Hence

$$\mathrm{MAut}(C) \cong C_{10} \times A_5$$

has order 600. For the displayed column representatives, the subgroup of pure coordinate permutations is trivial.

Proposition 3.4 (One orbit on received-word deep holes)**.** *The monomial automorphism group of* C *is transitive on the* 120 *deep-hole cosets. Let* Γ *be the group generated by these automorphisms and by translations* $v \mapsto v + c$ *for* $c \in C$*. Then*

$$|\Gamma| = |C|\,|\mathrm{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ *is transitive on all* 159720 *received-word deep holes. The stabilizer of each such word is cyclic of order five.*

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^{\times}$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w, choose $m \in \mathrm{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w. Both kinds of generator are Hamming isometries preserving C. The monomial group normalizes the translation subgroup, since $m\, t_c\, m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \mathrm{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. □

3.2 Decoding consequences

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters.

The special geometry also makes distance diagnosis and nearest-word structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$.

Proposition 3.5 (Complete syndrome oracle and ambiguity enumerator)**.** *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^{\mathsf{T}}$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 *nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are*

$$960, \qquad 150, \qquad 100, \qquad 120.$$

For a distance-two syndrome the number of nearest codewords is the number of secants through its direction, called its secant index. *Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.*

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 3.2 identifies the last set with $Q = 0$, proving the oracle. In the notation of its proof, the chord-intersection equations give $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$, later recognized as the H_3 strata in Section 6.1, where n_i counts projective directions of secant index i and n_0 is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. □

3.3 Self-polar structure

Edge's finite-plane construction [6, §§30.2–31] and Dye's general-field synthetic theory [15, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into a *synthematic total*, that is, a partition of the edges of K_6 into five perfect matchings. Section 5 turns its ten pairs of matchings into the Brianchon–support dictionary.

4 The rigidity theorem

Proposition 3.2 concerns one known arc. The following theorem is the paper's main claim: among *all* six-arcs of $\mathrm{PG}(2,11)$, the property "the projective deep-hole syndrome locus lies on a conic" picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

Remark 4.1 (Two conditions, one word apart). The condition studied here is that $\mathcal{U}(A)$ — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are logically unrelated: they constrain different point sets, and the arcs satisfying them are disjoint. Indeed every six-arc lying *on* a conic in $\mathrm{PG}(2,11)$ has $|\mathcal{U}(A)| \in \{18, 19, 20, 22\}$, so by (iii) none of them comes anywhere near the condition of the theorem.

We first isolate the geometric input that removes degenerate conics from the computer-assisted part of the argument.

The external-line case is a six-point instance of the nonexistence in odd order of non-trivial *hyperfocused arcs*, whose secants determine only $k-1$ points on an external line [3, 7, 19]. We give a direct proof because the three possible line positions are then handled together.

Lemma 4.2 (Six-arc line bound)**.** *Let A be a six-arc in $\mathrm{PG}(2,q)$, where q is odd. Every line contains at most $q-5$ points of $\mathcal{U}(A)$.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points: their intersections give a proper edge-colouring of K_5, whose chromatic index is five. Together with the vertex, at least six of the $q+1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \varnothing$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them, because their endpoint pairs are disjoint. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6, necessarily a one-factorization. After relabelling, three factors are

$$01|23|45, \qquad 02|14|35, \qquad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0 = (0,0)$, $p_1 = (x,0)$ and $p_2 = (0,y)$, where $xy \neq 0$. Writing $p_3 = (a,y)$, the first two factors force $p_4 = (x,b)$ and $p_5 = (a,b)$. Parallelism of 03 with 15 and with 24 gives two determinant equations whose difference is $2xy = 0$, impossible for odd q. Thus a disjoint line has at least six covered points, and in all cases at most $(q+1) - 6 = q-5$ points of $\mathcal{U}(A)$ remain. □

Theorem 4.3 (Rigidity theorem)**.** *For a six-arc $A \subset \mathrm{PG}(2,11)$, the following are equivalent:*

(i) the projective deep-hole syndrome locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) $|\mathcal{U}(A)| \leq 15$ (in fact, $|\mathcal{U}(A)| = 12$);

(iv) A is $\mathrm{PGL}(3, 11)$-equivalent to the Clebsch hexagon;

(v) the stabilizer of A in $\mathrm{PGL}(3, 11)$ contains A_5.

Proof. Suppose first that $\mathcal{U}(A)$ lies on a conic Q. If Q is nonsingular then $|\mathcal{U}(A)| \leq q+1 = 12$. If Q is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 4.2 again gives $|\mathcal{U}(A)| \leq 12$. On the other hand, the chord-defect identity of Lemma 6.1 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon A. Over a field of characteristic different from two, Dye proves that a hexagon has at most ten Brianchon points and that the equality cases form a single $\mathrm{PGL}_3(K)$-orbit [15, §2.2–2.3, Theorem 1, pp. 275–276]. Hence $c(A) \leq 10$, so $|\mathcal{U}(A)| \geq 12$; equality holds throughout, $c(A) = 10$, and A is a Clebsch hexagon. This proves (i)⇒(iv). Proposition 3.2 and projective equivariance give (iv)⇒(ii)⇒(i). Dye's stabilizer theorem gives (iv)⇒(v), and Storme and Van Maldeghem's finite-plane classification gives the converse.

It remains only to connect the numerical condition (iii). Any ordered four points of a six-arc form a projective frame, uniquely normalized to $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$. Enumerating the remaining unordered pair subject to the arc condition gives 1548 normalized sets. The least of the 360 ordered-frame normalizations is a complete projective-class key, giving fifteen classes; a class with stabilizer G occurs $360/|G|$ times. The resulting extension counts are

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\}.$$

Several of the fifteen projective classes share an extension count. Across the 1548 frame-normalized representatives, the corresponding multiplicities are

$$(6, 30, 150, 300, 630, 360, 72).$$

The six representatives attaining 12 form the single Clebsch class; every other class has at least 16 extension points. Thus (iii)⇔(iv), completing the equivalence. The exhaustive calculation is therefore used for the size-gap clause, not for the conic containment implication. □

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 4.4 (Low-degree algebraic rigidity)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q, and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

Computer-assisted proof. For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on $\mathcal{U}(A)$. The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is generated by Q, and multiplication by linear forms gives the full three-dimensional cubic kernel. □

Corollary 4.5 (Monomial characterization)**.** *Up to monomial equivalence, C is the unique linear $[6,3,4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. In particular, A_5 is recovered from this coding condition, not assumed as a hypothesis.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 4.4 makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and the stabilizer statement is clause (v) of the theorem. □

Remark 4.6 (Sharpness of the degree threshold)*.* The cutoff cannot be raised from three to four: one non-Clebsch class has an 18-point uncovered locus equal to the rational points of a smooth absolutely irreducible quartic of genus three.

The rigidity theorem has quantitative consequences at three scales: the number of deep-hole directions, the distance of their locus from the nearest conic, and the effect of replacing one vertex of the hexagon.

For the second, let $\mathfrak{C}$ be the set of nonsingular conics in $\mathrm{PG}(2,11)$ and define the *nearest-conic discrepancy*

$$\delta(A) = \min_{\mathcal{Q}\in\mathfrak{C}} |\mathcal{U}(A) \triangle \mathcal{Q}(\mathbb{F}_{11})|.$$

This is a PGL-invariant function, not a metric on arcs: equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A)$, while PGL permutes $\mathfrak{C}$.

For the local statement, call two embedded six-arcs *neighbors* when one is obtained from the other by replacing one point. Fix the displayed Clebsch arc A and its standard conic $\mathcal{C}$, and put

$$d_{\mathcal{C}}(B) = |\mathcal{U}(B) \triangle \mathcal{C}|$$

for every embedded six-arc B.

Theorem 4.7 (Quantitative gaps)**.** *(a) There is no six-arc $A \subset \mathrm{PG}(2,11)$ with $12 < |\mathcal{U}(A)| < 16$. Every arc attaining $|\mathcal{U}(A)| = 12$ is Clebsch up to projectivity. If $|\mathcal{U}(A)| \geq 16$, then $\mathcal{U}(A)$ lies on no conic.*

(b) Every non-Clebsch class satisfies $\delta(A) \geq 12$, and the bound is sharp.

(c) The arc A has 252 distinct neighbors. Every neighbor B satisfies $d_{\mathcal{C}}(B) \geq 18$ and $|\mathcal{U}(B) \cap \mathcal{C}| \leq 7$; none has its uncovered locus contained in a conic, degenerate or otherwise.

The first distribution below counts projective classes. The next two count embedded one-point replacements of the fixed arc A; the final list then groups those replacements into A_5-orbits:

$$\begin{aligned}
\#\{[B] : \delta(B) = d\} &= \{0{:}1, 12{:}2, 13{:}1, 14{:}3, 15{:}1, 16{:}4, 17{:}2, 18{:}1\}, \\
\#\{B \sim A : d_{\mathcal{C}}(B) = d\} &= \{18{:}30, 19{:}60, 20{:}90, 22{:}42, 24{:}30\}, \\
\#\{B \sim A : |\mathcal{U}(B) \cap \mathcal{C}| = m\} &= \{4{:}60, 6{:}132, 7{:}60\}.
\end{aligned} \tag{1}$$

Under A_5, the neighbors split into eight orbits with (orbit size, discrepancy)

$$(30, 18), (60, 19), (30, 20), (30, 20), (30, 20), (12, 22), (30, 22), (30, 24).$$

Remark 4.8 (Global versus fixed-conic discrepancy). The local bound is tied to the fixed pair $(A, \mathcal{C})$. Indeed, the conic-preserving projectivity $g : (X, Y, Z) \mapsto (4X, 2Y, Z)$ sends A to a distinct embedded six-arc, while equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A) = \mathcal{C}$. Hence $d_{\mathcal{C}}(gA) = 0$ although $gA \neq A$. The global statement therefore minimizes over conics and passes to projective classes; the bound eighteen applies only to one-point replacements of the fixed arc.

Computer-assisted proof. The first clause restates Theorem 4.3. For the second, the enumeration normalizes all 1548 representatives to fifteen projective classes, enumerates the 160930 nonsingular conics, and computes the first distribution in (1).

For the third, it deletes each vertex in turn, tests every replacement, and obtains 42 choices per vertex and 252 distinct neighbors. Exact quadratic-monomial evaluation matrices have full rank six, so no nonzero quadratic vanishes on a neighbor's uncovered locus. The remaining two distributions and the eight A_5-orbits follow from the sixty stabilizing projectivities. $\square$

Remark 4.9 (Why this rigidity is not in the extension-locus literature). The set $\mathcal{U}(A)$ is the classical extension locus, studied for sufficiently large arcs through Segre's tangent-envelope method; see Ball and Lavrauw [11, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime, while Sadeh's census uses six-arcs for a different classification problem. Neither direction records the conic rigidity detected here.

5 The chirality bipartition

Label the six coordinates by $0, \ldots, 5$. The 20 three-element coordinate supports form two complementary A_5-orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive: this unoriented two-class structure is what we call *chirality*. Its ten complementary pairs carry the Petersen graph in Figure 1.

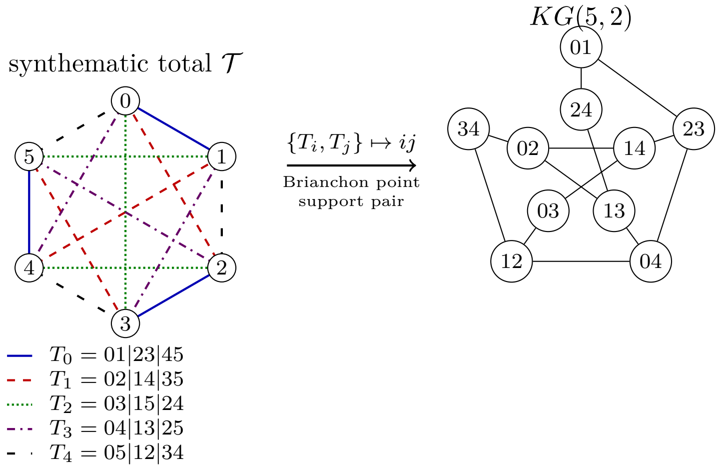

Figure 1: The synthematic–Petersen dictionary. The five line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair $\{T_i, T_j\}$, their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex ij; adjacency means that the index pairs are disjoint.

Proposition 5.1 (Brianchon–support dictionary)**.** *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [6, 15], regarded as the five perfect matchings in their synthematic total, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle. Let $\beta(T, T') = \{S, S^{\mathsf{c}}\}$ be the unordered pair of alternating vertex classes of the cycle. For example, $T_0 \cup T_1$ is the cycle $0, 1, 4, 5, 3, 2$, so $M(T_0, T_1) = 05|13|24$ and $\beta(T_0, T_1) = \{\{0, 3, 4\}, \{1, 2, 5\}\}$.*

The three chords indexed by $M(T, T')$ concur at a point $b(T, T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \qquad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^{\mathsf{c}}\} : |S| = 3\}, \qquad \{T, T'\} \longmapsto \beta(T, T'),$$

are A_5-equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of the synthematic total. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten perfect matchings outside the total. Applying the alternating-class rule to the five displayed matchings gives ten distinct complementary support pairs, hence all ten such pairs. Edge's Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [6, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the invariant total and incidence, both bijections and their compatibility with the support map are A_5-equivariant. □

Corollary 5.2 (The decoder reconstructs the Brianchon configuration). *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five-matching synthematic total.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 5.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings complementary to the total. □

Proposition 5.3 (Invariant support chirality). *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary A_5-orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s, each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200 + 1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The other coset in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 5.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.

Proof. The degree-six action of A_5 on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique synthematic total, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 1 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome s and a three-support S. The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s. None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A, contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S. Finally, every monomial automorphism induces a support permutation in A_5, by the exact sequence in Section 3.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve C, even after coordinate rescaling, because its support permutations lie outside that quotient A_5. □

6 Why $q = 11$

Call an arc A *conic-filling* if $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{Q}$. The conic-filling syndrome phenomenon of Sections 3–4 is specific to $q = 11$, even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity makes this a four-field question, and the only nontrivial residue is a classical exact-distance problem in the Sylvester graph.

Lemma 6.1 (Chord-defect identity). *Let A be a six-arc in* $\mathrm{PG}(2, q)$*, and let $c(A)$ be the number of points off A incident with three of its fifteen chords. Then*

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad 0 \le c(A) \le 15.$$

Consequently, if $|\mathcal{U}(A)| = q + 1$, then

$$c(A) = (q - 6)(q - 9).$$

Proof. At a point off A, concurrent chords have pairwise disjoint endpoint pairs, so at most three chords occur. Let n_i count the off-arc points on exactly i chords. Counting chord–point incidences and pairs of chords with disjoint endpoints gives

$$n_1 + 2n_2 + 3n_3 = 15(q-1), \qquad n_2 + 3n_3 = \binom{15}{2} - 6\binom{5}{2} = 45.$$

Writing $c(A) = n_3$, the number of covered points off A is $n_1 + n_2 + n_3 = 15q - 60 + c(A)$. Subtracting this from the q^2+q-5 points off A proves the identity. Each triple concurrence uses one of the fifteen perfect matchings of the six endpoints, and a fixed matching can be concurrent at only one point; hence $c(A) \leq 15$. The last display follows on setting $|\mathcal{U}(A)| = q+1$. □

Theorem 6.2 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2,q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. Moreover, at $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Lemma 6.1 gives $c(A) = (q-6)(q-9)$ with $0 \leq c(A) \leq 15$. The prime-power condition therefore leaves only $q \in \{4,5,9,11\}$: the values at 7 and 8 are negative, while every integer $q \geq 12$ gives $c(A) \geq 18$.

At $q = 4$, every six-arc is a hyperoval. Each point off it lies on three secants (count the six hyperoval points on the five lines through the point), so $\mathcal{U}(A)$ is empty. At $q = 5$, every six-arc is an oval and hence a nonsingular conic; the standard internal/external point counts show that every off-conic point lies on a secant, so again $\mathcal{U}(A)$ is empty.

It remains to exclude $q = 9$. The argument has three steps: the six arc points must be internal to $\mathcal{C}$; passant joins of internal points become distance-two pairs in the Sylvester graph; and that distance graph has clique number five. Since every point of $\mathcal{C}$ is uncovered, no chord of A meets $\mathcal{C}$; every chord is therefore passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external; hence the five chords through each vertex force all six vertices of A to be internal. Let Σ be the graph on the 36 internal points, with $P \sim Q$ when $Q \in P^\perp$ under the conic polarity. Standard conic counts give the intersection array

$$\{5,4,2;1,1,4\},$$

so Σ is the Sylvester graph [13, §13.1.2]; see also the combinatorial uniqueness proof in [14, Thm. 6]. For distinct internal points, the unique possible common neighbor is $P^\perp \cap Q^\perp = (PQ)^\perp$; consequently PQ is passant exactly when P, Q are at distance two in Σ. Thus A would be a six-clique in the graph joining pairs at distance exactly two in Σ. But its clique number, denoted $\mathrm{eq}_2(\Sigma)$ in [12, Table 1], is five, a contradiction.

Therefore $q = 11$. The final assertion follows from Theorem 4.3, which puts every matching $q = 11$ representative in the single Clebsch orbit. □

Proposition 6.3 (The Clebsch-family uncovered formula)**.** *Let K be a Clebsch hexagon over $\mathbb{F}_q$. Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod 4$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$ and

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q-4)(q-11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [15, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(K)$. Thus $c(K) = 10$, and Lemma 6.1 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of K are non-secants of the associated conic when $q \equiv 3 \pmod 4$ [15, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points. □

Remark 6.4 (The next admissible field). At $q = 19$, the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [15, p. 281], while Proposition 6.3 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus $120 = (19 - 4)(19 - 11)$ additional points escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at $q = 11$.

6.1 The two reflection-arrangement exceptions

The chord moments below classify the neighboring arc sizes. The two survivors have a common arrangement-theoretic organization, although the arrangement calculation alone gives complement sizes, not conic equations.

Let F be a field of characteristic different from two containing τ with $\tau^2 = \tau + 1$. In dual coordinates, take the fifteen line normals

$$(1,0,0), (0,1,0), (0,0,1), \qquad \operatorname{cyc}(1, \epsilon\tau, \delta(\tau - 1)) \quad (\epsilon, \delta \in \{\pm 1\}).$$

Over the real embedding with $\tau = (1 + \sqrt{5})/2$, their signed versions are the roots of type H_3; over other fields this denotes the displayed finite reduction of the icosahedral mirror arrangement.

Proposition 6.5 (The Clebsch secants as reduced H_3 mirrors)**.** *The six fivefold points are*

$$(0, 1, 1 - \tau), (0, 1, \tau - 1), (1, 1 - \tau, 0), (1, \tau - 1, 0), (1, 0, -\tau), (1, 0, \tau).$$

They form an arc, their fifteen joins are the mirrors, and the other singularities are ten triple and fifteen double points. Over $\mathbb{F}_{11}$, take $\tau = 8$. The matrix

$$T = \begin{pmatrix} 2 & 3 & 8 \\ 10 & 6 & 9 \\ 2 & 2 & 5 \end{pmatrix}$$

maps these six column vectors onto the displayed Clebsch hexagon A. Writing line coefficients as rows, the induced map $\ell \mapsto \ell T^{-1}$ carries the mirrors onto the secants of A.

Proof. Pairwise cross-products of the normals give the stated $6_5, 10_3, 15_2$ ledger, and joining the six fivefold points recovers all fifteen normals. Every three-by-three determinant on the six points is twice a unit in $\mathbb{Z}[\tau]$, so the set is an arc when $\operatorname{char} F \neq 2$. Finally, $\det T = 3$ in $\mathbb{F}_{11}$ and direct multiplication gives the two asserted set equalities. The exact replay is `check_reflection_arrangements.py`. □

Characteristic two is bad because the signs coalesce. In every other characteristic containing τ, the displayed lattice survives; this includes the lattice-faithful but nonsemisimple characteristic-five reduction. For the central rank-three hyperplane arrangement, a point where m projective mirrors meet has Möbius value $m-1$. Thus

$$\chi_{H_3}(t) = t^3 - 15t^2 + 59t - 45 = (t-1)(t-5)(t-9).$$

Dividing the affine complement count by $q-1$ gives $(q-5)(q-9)$ projective points. At $q = 11$, the points lying on respectively zero, one, two, and three mirrors number $12, 90, 15, 10$; here "one" means an ordinary mirror point. With the six fivefold points marked as columns, these are exactly the projective strata behind Proposition 3.5; the general passage to coset-leader multiplicities is due to Jurrius and Pellikaan [18].

For the standard four-frame $E = \{e_1, e_2, e_3, (1,1,1)\}$, the six joins have equations

$$X, Y, Z, X-Y, X-Z, Y-Z = 0.$$

These are the essentialized braid mirrors $x_i = x_j$ of type A_3, after setting $x_4 = 0$. Their four triple and three double points give

$$\chi_{A_3}(t) = (t-1)(t-2)(t-3), \qquad |\mathrm{PG}(2,q) \setminus \bigcup A_3| = (q-2)(q-3).$$

In the faithful-reduction range just stated for the explicit H_3 model, the two size equations are therefore

arc	secant arrangement	$\lvert\mathcal{U}\rvert = q+1$
four-frame	A_3	$(q-1)(q-5) = 0$
Clebsch hexagon	H_3	$(q-4)(q-11) = 0.$

The extraneous root $q = 4$ lies precisely in the bad characteristic-two regime, where the displayed H_3 reduction degenerates. Thus the arrangements organize the surviving sizes $q = 5, 11$; the separate geometry that makes the complements nonsingular conics is Proposition 3.2 for the Clebsch case and the direct quadratic verification in the proof below for the frame.

Theorem 6.6 (Conic-filling loci for $4 \leq k \leq 7$)**.** *Let A be a k-arc in $\mathrm{PG}(2,q)$, where q is a prime power and $4 \leq k \leq 7$. If $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$-rational points of a nonsingular conic, then*

$$(k,q) = (4,5) \qquad \text{or} \qquad (k,q) = (6,11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur.

Proof with exhaustive finite verification. Let n_i count off-arc points incident with exactly i chords. For $k \leq 7$, at most three pairwise disjoint chords concur. Counting chord–point incidences and pairs of disjoint chords gives the two universal moments

$$\sum_i i n_i = \binom{k}{2}(q-1), \qquad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

For $k = 4$, every arc is a projective frame, so the A_3 calculation above gives $|\mathcal{U}(A)| = (q-2)(q-3)$. Equating this with $q+1$ gives $q = 5$; for the standard frame a direct substitution further gives

$$\mathcal{U}(A) = V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

where $V(f)$ denotes the projective zero set of f; this is a nonsingular six-point conic. For $k = 5$, the same moments force $n_2 = 15$ and $|\mathcal{U}(A)| = q^2 - 9q + 21$; equality with $q+1$ would require $q^2 - 10q + 20 = 0$, which has no integral root. The case $k = 6$ is Theorem 6.2 together with Theorem 4.3.

For $k = 7$, the moments give

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q+1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The standard arc-size bound gives $q \geq 7$. The condition $n_2 \geq 0$ gives $q \leq 15$, so the prime-power candidates are $q \in \{7, 8, 9, 11, 13\}$; then $n_1 \geq 0$ eliminates $7, 8, 9$. Thus only $q = 11$ and $q = 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized enumeration now supplies the finite exclusion. It finds 10232 distinct normalized seven-arcs at $q = 11$, of which 140 have $|\mathcal{U}(A)| = q+1$, and 53960 at $q = 13$, of which 1680 have that size. Every seven-arc is reached because it contains a four-frame; extending each normalized six-arc by its uncovered points produces every resulting seven-arc exactly three times, according to which non-frame point is deleted. For all 1820 candidates of the required size, the quadratic evaluation matrix has full rank six, so none lies on a conic. $\square$

For $k \geq 8$ the second moment leaves additional concurrency parameters, so the analogous classification remains open.

7 Verification architecture

Tables 1 and 2 separate mathematical input, kernel-checked formal proof, and exhaustive finite enumeration. Here "kernel-checked" means that Lean's trusted kernel checks the stated implication; executable replays instead enumerate the finite search space with exact arithmetic. Paths are relative to `lean/RelativeConicArcs/` and the paper directory.

Result	Mathematical input	Kernel-checked formalization
A_5 orbits and syndrome conic	Dye; Proposition 3.1	`Q11A5PointOrbits.lean`; `Q11Coding.lean`
A_3/H_3 arrangement synthesis	exact root coordinates; intersection lattices	—
Decoding and chirality	arc–coset dictionary; Edge–Dye triangles	`Q11DecodingSynthesis.lean`
Rigidity theorem	line bound; chord defect; Dye's Theorem 1	`Q11DyeAxioms.lean`; `Q11DyeConsequences.lean`; `SixArcDefectBridge.lean`
Gap and low-degree loci	finite linear algebra over $\mathbb{F}_{11}$	—
Uniqueness of $q = 11$	chord moments; cited Sylvester bound	`ClebschChordDefect.lean`; `Q9Sylvester.lean`
Clebsch-family formula	Dye's ten Brianchon points and edge criterion	—
The range $4 \leq k \leq 7$	two universal chord moments	`SmallKChordMoments.lean`; `SmallKGeometricBridge.lean`

Table 1: Mathematical and formal dependencies of the principal results.

Result	Exhaustive or independent executable check
A_5 orbits and syndrome conic	`check_rigidity_degenerate_conic.py`
A_3/H_3 arrangement synthesis	`check_reflection_arrangements.py`
Decoding and chirality	`check_decoding.py`; `check_chirality.py`; `check_code_automorphisms.py`
Rigidity theorem	1548-representative frame-normalized enumeration
Gap and low-degree loci	`check_global_conic_gap.py`; `check_perturbation_gap.py`; `check_low_degree_loci.py`; Singular calculation
Uniqueness of $q = 11$	`check_small_q_uniqueness.py`; independent $q = 9$ construction
Clebsch-family formula	`check_q19_nonexample.py`
The range $4 \leq k \leq 7$	`check_small_k_conic_filling.py`

Table 2: Executable checks, separate from the formal dependencies in Table 1.

The formal development axiomatizes exactly two consequences of Dye's Theorem 1: the ten-point Brianchon bound and the classification of equality. The line bound and chord-defect identity are proved independently. At $q = 11$, the finite census also checks the imported numerical conclusion: $|\mathcal{U}(A)| = 12 = 22 - 10$ is equivalent to the maximal value $c(A) = 10$.

7.1 Independent small-field census

Normalizing an ordered four-frame and enumerating the remaining unordered pair gives the following exact check.

q	representatives	histogram of $\lvert\mathcal{U}(A)\rvert$	conic matches
2	0	—	0
3	0	—	0
4	1	0^1	0
5	3	0^3	0
7	70	$0^{40}, 2^{30}$	0
8	195	$0^{45}, 4^{150}$	0
9	441	$0^6, 4^{15}, 6^{120}, 7^{120}, 8^{180}$	0
11	1548	$12^6, 16^{30}, 18^{150}, 19^{300}, 20^{630}, 21^{360}, 22^{72}$	6
13	4015	$36^{85}, 38^{210}, 39^{480}, 40^{1080}, 41^{1800}, 42^{360}$	0

Table 3: Small-field census. "Representatives" means frame-normalized point sets, not projective classes; m^r means that r representatives have $|\mathcal{U}(A)| = m$. The six conic matches at $q = 11$ form one projective class.

The enumeration uses exact arithmetic in every field and tests both nonsingular and singular conic forms in characteristic two. A separate $q = 9$ certificate constructs the polarity graph, verifies the passant/distance-two dictionary, and computes clique number five.

8 Conclusion

The Clebsch code shows that covering-radius data can recover geometry that was not built into the code: conic containment of the deep-hole directions forces the Clebsch hexagon and hence its A_5 symmetry. The proof separates the mechanism from the census. A line bound and a chord-defect identity reduce the geometric condition to Dye's equality case; enumeration is needed only for the sharp numerical gap and its refinements.

The same chord moments explain both the isolation of $q = 11$ and the classification for $4 \leq k \leq 7$. For $k \geq 8$, further concurrency moments are needed. A second problem is to connect fixed-k rigidity with the large-arc conic-forcing regime. Both ask how far a syndrome locus can reconstruct the projective system that produced it.

References

[1] C. A. Athanasiadis, Characteristic polynomials of subspace arrangements and finite fields, *Advances in Mathematics* **122** (1996), 193–233. doi:10.1006/aima.1996.0059.

[2] A. Blokhuis, Á. Seress, and H. A. Wilbrink, Characterization of complete exterior sets of conics, *Combinatorica* **12**(2) (1992), 143–147. doi:10.1007/BF01204717.

[3] A. Bichara and G. Korchmáros, Note on $(q+2)$-sets in a Galois plane of order q, in *Combinatorial and Geometric Structures and Their Applications*, Annals of Discrete Mathematics **14**, North-Holland, 1982, 117–121. doi:10.1016/S0304-0208(08)73588-6.

[4] S. Calvo, The icosahedral line configuration and Waldschmidt constants, *Journal of Pure and Applied Algebra* **228**(6) (2024), 107563. doi:10.1016/j.jpaa.2023.107563; arXiv:2209.01499.

[5] A. Clebsch, Ueber die Anwendung der quadratischen Substitution auf die Gleichungen 5-ten Grades und die geometrische Theorie des ebenen Fünfseits, *Mathematische Annalen* **4** (1871), 284–345.

[6] W. L. Edge, Conics and orthogonal projectivities in a finite plane, *Canadian Journal of Mathematics* **8** (1956), 362–382.

[7] D. A. Drake and K. Keating, Ovals and hyperovals in Desarguesian nets, *Designs, Codes and Cryptography* **31** (2004), 195–212. `https://doi.org/10.1023/B:DESI.0000015883.89954.80`; arXiv:math/0202115.

[8] T. Rudd, Arcs complete outside a prescribed conic: An exact defect identity and $\rho_{\mathcal{C}}(16) = 9$, working paper, 2026.

[9] Q. Cheng and E. Murray, On deciding deep holes of Reed–Solomon codes, in *Theory and Applications of Models of Computation (TAMC 2007)*, Lecture Notes in Computer Science **4484**, Springer, 2007, 296–305.

[10] A. A. Davydov, S. Marcugini, and F. Pambianco, On the weight distribution of the cosets of MDS codes, *Advances in Mathematics of Communications* **17**(5) (2023), 1115–1138. arXiv:2101.12722v2. (Numbered results are cited by their arXiv v2 numbering.)

[11] S. Ball and M. Lavrauw, Arcs in finite projective spaces, *EMS Surveys in Mathematical Sciences* **6** (2019), 33–45. arXiv:1908.10772.

[12] A. Abiad, A. Jabal Ameli, and L. Reijnders, The clique number of the exact distance t-power graph: complexity and eigenvalue bounds, *Discrete Applied Mathematics* **363** (2025), 55–70. doi:10.1016/j.dam.2024.11.032.

[13] A. E. Brouwer, A. M. Cohen, and A. Neumaier, *Distance-Regular Graphs*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 18, Springer-Verlag, Berlin, 1989.

[14] A. Jurišić and J. Vidali, The Sylvester graph and Moore graphs, *European Journal of Combinatorics* **80** (2019), 184–193. doi:10.1016/j.ejc.2018.02.018.

[15] R. H. Dye, Hexagons, conics, A_5 and $\mathrm{PSL}_2(K)$, *Journal of the London Mathematical Society* (2) **44** (1991), 270–286. doi:10.1112/jlms/s2-44.2.270.

[16] V. Guruswami and A. Vardy, Maximum-likelihood decoding of Reed–Solomon codes is NP-hard, *IEEE Transactions on Information Theory* **51**(7) (2005), 2249–2256.

[17] J. W. P. Hirschfeld, *Projective Geometries over Finite Fields*, second edition, Oxford University Press, 1998.

[18] R. Jurrius and R. Pellikaan, The coset leader and list weight enumerator, in *Topics in Finite Fields*, Contemporary Mathematics **632**, American Mathematical Society, 2015, 229–251. doi:10.1090/conm/632/12631.

[19] S.-L. Ng and P. R. Wild, On k-arcs covering a line in finite projective planes, *Ars Combinatoria* **58** (2001), 289–300.

[20] A. R. Sadeh, *Classification of k-arcs in* $\mathrm{PG}(2, 11)$ *and cubic surfaces in* $\mathrm{PG}(3, 11)$, D.Phil. thesis, University of Sussex, 1984.

[21] L. Storme and J. A. Thas, Generalized Reed–Solomon codes and normal rational curves: an improvement of results by Seroussi and Roth, in *Advances in Finite Geometries and Designs*, Oxford University Press, 1991, 369–389.

[22] L. Storme and H. Van Maldeghem, Primitive arcs in $\mathrm{PG}(2, q)$, *Journal of Combinatorial Theory, Series A* **69** (1995), 200–216. doi:10.1016/0097-3165(95)90051-9.

[23] Y. Zhang, D. Wan, and K. Kaipa, Deep holes of projective Reed–Solomon codes, *IEEE Transactions on Information Theory* **66**(4) (2020), 2392–2401. arXiv:1901.05445.